List Recovery of Reed–Solomon Subcodes over Subfield Evaluation Points

A proof of Theorem 28

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Abstract

We prove the list-recovery extension of the Guruswami–Xing construction: for fixed input-list size ℓ, explicit linear subcodes of Reed–Solomon codes at arbitrary distinct subfield evaluation points have all messages with agreement at least $R+\varepsilon$ contained in a linear space of dimension $O(\ell/\varepsilon^2)$. The proof combines list interpolation, a coefficientwise Frobenius recurrence, and an explicit strong subspace design. We give all reductions and rounding details; the sole external construction theorem is the strong subspace design of Guruswami and Kopparty.

1 Precise statement and conventions

An $\mathbb{F}_q$-linear code $C \subseteq (\mathbb{F}_{q^s})^n$ of dimension k has rate $k/(sn)$. A table is a tuple $T = (T_1, \dots, T_n)$ with $T_i \subseteq \mathbb{F}_{q^s}$ and $|T_i| \le \ell$. Agreement means the number of indices i for which $c_i \in T_i$. In particular, our assertion also applies to tables with lists of size exactly ℓ.

Theorem 1. *Let $\ell \ge 1$ be an integer and $R, \varepsilon \in (0,1)$. For every prime power $q \ge n \ge 1$ and distinct $\alpha_1, \dots, \alpha_n \in \mathbb{F}_q$, there is an explicit $\mathbb{F}_q$-linear subcode C of a Reed–Solomon code over $\mathbb{F}_{q^s}$ on these points, with*

$$s = O(\ell/\varepsilon^2), \qquad k = \lfloor Rsn \rfloor, \qquad \frac{k}{sn} \in \left(R - \frac{1}{sn}, R\right].$$

For every table T, the messages of codewords with agreement at least $(R+\varepsilon)n$ lie in a linear subspace of $\mathbb{F}_q^k$ of dimension $O(\ell/\varepsilon^2)$. A basis of such a containing subspace can be computed in polynomial time in $q, n, \ell, 1/\varepsilon$, given a representation of the finite fields.

For fixed ℓ the extension degree is $O_\ell(1/\varepsilon^2)$, as in the supplied theorem with constant ℓ. The proof makes its ℓ-dependence explicit. It does not assert an extension-degree bound with an absolute constant independent of ℓ. Exact rate R is possible when Rsn is integral; for arbitrary real R, the displayed rounding is necessary. One can instead take $k = \lceil Rsn \rceil$ in the construction below, obtaining rate at least R and less than $R + 1/(sn)$, with the same agreement threshold.

2 The explicit construction ingredient

A strong (r, A) subspace design in $\mathbb{F}_q^m$ is a family of linear spaces H_i such that

$$\sum_i \dim_{\mathbb{F}_q}(W \cap H_i) \le A \quad \text{for every } W \subseteq \mathbb{F}_q^m, \quad \dim_{\mathbb{F}_q} W \le r.$$

The following convenient form follows from the main construction theorem of Guruswami and Kopparty [1, Theorem 7].

Lemma 2. *There is an absolute constant K such that for $0 < \eta < 1$, $r \geq 1$, and $m = \lceil Kr/\eta \rceil$, whenever $q > m$, one can explicitly construct at least q spaces $H_i \subseteq \mathbb{F}_q^m$ of codimension at most ηm forming a strong (r, m) subspace design. Construction time is polynomial in q, m.*

Proof. The cited theorem, with target dimension r, multiplicity parameter $2r$, and irreducible-degree parameter 3, gives $M \geq cq^3/(6r)$ spaces of codimension $6r$, and intersection bound

$$A = \frac{(m-1)r}{3(r+1)} \leq m.$$

Here $c > 0$ is absolute. Choose $K \geq 6$ large enough that $cK^2/6 \geq 1$, also increasing it for the absolute constant implicit in the construction. Then $6r \leq \eta m$, $2r \leq m < q$, and $cq^2/(6r) \geq 1$ since $q > m \geq Kr$. Hence $M \geq q$. Take any q of the spaces. Bounds for dimension smaller than r follow by extending the tested space to dimension r. The cited construction is polynomial time for this fixed irreducible degree. □

3 Interpolation and the coefficient recurrence

Fix $E = \mathbb{F}_{q^s}$ and write $\sigma(\sum_j f_j X^j) = \sum_j f_j^q X^j$. Because $\alpha \in \mathbb{F}_q$,

$$\sigma^a(f)(\alpha) = f(\alpha)^{q^a}. \tag{1}$$

Lemma 3. *Let $1 \leq d \leq n$, $h \leq s$, and let t be an integer with $d \leq t \leq n$. If*

$$t + h(t - d + 1) > \ell n, \tag{2}$$

then for every table T there are polynomials $A, B_0, \ldots, B_{h-1} \in E[X]$, not all zero, with

$$\deg A < t, \qquad \deg B_a < t - d + 1,$$

for which every $f \in E[X]$ of degree less than d with agreement at least t satisfies

$$A + \sum_{a=0}^{h-1} B_a \sigma^a(f) = 0. \tag{3}$$

At least one B_a is nonzero.

Proof. For each i and $y \in T_i$ impose $A(\alpha_i) + \sum_a B_a(\alpha_i) y^{q^a} = 0$. These are at most ℓn homogeneous linear equations over E in $t + h(t - d + 1)$ unknown coefficients. Thus a nonzero solution exists and is obtained by linear algebra. The left side of (3) has degree less than t. At every agreeing coordinate it vanishes by (1), so at least t distinct roots force it to be zero. If all B_a were zero, a nonzero A would vanish at every coordinate with a nonempty list. If those lists are all nonempty, this contradicts $\deg A < t \leq n$. For tables with empty lists, first enlarge every list to size ℓ (or to size $\min\{\ell, |E|\}$). This only enlarges the candidate set, and preserves the equation-count bound. Use this enlarged table throughout. □

Lemma 4. *Assume at least one $B_a \neq 0$ and $h \leq s$. The solution set $V \subseteq E^d$ to (3), identified with coefficient tuples, is empty or an affine $\mathbb{F}_q$-space. There is a fixed linear space $W \subseteq E$ of dimension at most $h - 1$ such that, after any prefix $f_0, \ldots, f_{j-1}$ is fixed, the possible next coefficients f_j lie in a coset of W.*

Proof. All equations are affine linear over $\mathbb{F}_q$. Let $v = \min_a \operatorname{ord}_X B_a$ and write $b_a = [X^v]B_a$. The polynomial

$$L(Z) = \sum_{a=0}^{h-1} b_a Z^{q^a}$$

is nonzero and $\mathbb{F}_q$-linear as a map $E \to E$. Its degree is at most q^{h-1}, so its kernel W has at most q^{h-1} elements, hence dimension at most $h-1$. The coefficient of X^{v+j} in (3) gives

$$L(f_j) = -[X^{v+j}]A - \sum_{a=0}^{h-1} \sum_{u=0}^{j-1} [X^{v+j-u}]B_a \, f_u^{q^a}.$$

The right side depends only on the prefix. Each nonempty fiber of L is a coset of W. This argument uses no translation of the evaluation variable, and therefore applies without any restriction on the characteristic or on whether 0 is an evaluation point. □

Lemma 5. *Let $V \subseteq E^d$ be an affine space with the prefix property in Lemma 4, with $\dim_{\mathbb{F}_q} W \le r$. Identify E with $\mathbb{F}_q^s$, and let $H_0, \dots, H_{d-1}$ be a strong (r, A_*) subspace design. Put $U = \prod_{j=0}^{d-1} H_j$. Then $V \cap U$ is empty or an affine space of dimension at most A_*. Its linear span has dimension at most $A_* + 1$.*

Proof. Suppose the intersection is nonempty and write it as $z + D$. For each j, let D_j be the image of D on its first j coordinates, so $D_0 = 0$ and $D_d = D$. The kernel of $D_{j+1} \to D_j$, identified with its last coordinate, is a subspace of $W \cap H_j$: its vectors have zero earlier coordinates, and their jth coordinates are differences of two allowed continuations. Consequently

$$\dim_{\mathbb{F}_q} D = \sum_{j=0}^{d-1} (\dim_{\mathbb{F}_q} D_{j+1} - \dim_{\mathbb{F}_q} D_j) \le \sum_{j=0}^{d-1} \dim_{\mathbb{F}_q} (W \cap H_j) \le A_*.$$

Finally $z + D \subseteq \operatorname{span}_{\mathbb{F}_q}(D \cup \{z\})$, of dimension at most $A_* + 1$. □

4 Construction and proof of the theorem

Proof of Theorem 1. If $R + \varepsilon > 1$, the assertion is vacuous; take $s = 1$ and any dimension-$\lfloor Rn \rfloor$ subcode of the full length-n Reed–Solomon code. Hence assume $R + \varepsilon \le 1$. Set

$$\eta = \varepsilon/8, \qquad h = \lceil 4\ell/\varepsilon \rceil, \qquad r = h - 1, \qquad m = \lceil Kr/\eta \rceil,$$

where the absolute constant K is as in Lemma 2, increased if needed so that $m \ge h$ and $m \ge 8/\varepsilon$. In particular $m = O(\ell/\varepsilon^2)$.

Small cases. If $q \le m$ or $n < 8/\varepsilon$, use $s = 1$ and any $\lfloor Rn \rfloor$-dimensional subcode of the full Reed–Solomon code of degree less than n. Its entire message space has dimension at most m, so it itself is a containing space for every table. This handles all small fields and lengths without an asymptotic qualification.

Main case. Suppose $q > m$ and $n \ge 8/\varepsilon$. Use extension degree $s = m$ and put

$$d = \lfloor (R + \varepsilon/2)n \rfloor, \qquad t = \lceil (R + \varepsilon)n \rceil.$$

We have $1 \le d \le t \le n$, $t - d \ge \varepsilon n/2$, and

$$t + h(t - d + 1) > h\varepsilon n/2 \ge 2\ell n > \ell n.$$

Choose d spaces $H_0, \dots, H_{d-1} \subseteq E \cong \mathbb{F}_q^m$ from Lemma 2. This is possible since $d \le n \le q$. Restrict the coefficient tuple of a degree-less-than-d polynomial to

$$U = H_0 \times \cdots \times H_{d-1}.$$

Evaluation is injective since $d \le n$ and the evaluation points are distinct. Thus its image is an $\mathbb{F}_q$-linear Reed–Solomon subcode of dimension at least $(1-\eta)md$. Moreover

$$\frac{d}{n} \ge R + \frac{\varepsilon}{2} - \frac{1}{n} \ge R + \frac{3\varepsilon}{8}, \qquad (1-\eta)\frac{d}{n} \ge \frac{d}{n} - \eta \ge R + \frac{\varepsilon}{4}.$$

Here $d/n \le 1$ was used in the second inequality. Select by linear algebra any subspace $U' \subseteq U$ of dimension $k = \lfloor Rmn \rfloor$, and let C be its evaluation image. Fix an $\mathbb{F}_q$-linear isomorphism $\phi : \mathbb{F}_q^k \to U'$ as the message parametrization. The positive dimension slack also permits $k = \lceil Rmn \rceil$ if the alternative rounding convention is desired.

Given a table, enlarge empty lists as in Lemma 3. That lemma supplies (3), and every qualifying message polynomial satisfies it. Its solution space V has the prefix property of Lemma 4. Lemma 5 and the strong (r, m) design imply that $V \cap U$ is empty or has affine dimension at most m. Since $U' \subseteq U$, the same holds for $V \cap U'$. The preimage $\phi^{-1}(V \cap U')$ therefore lies in a linear subspace of $\mathbb{F}_q^k$ of dimension at most

$$m + 1 = O(\ell/\varepsilon^2).$$

If it is empty, use the zero subspace. This proves the required linear, rather than merely affine, containment.

All choices of coefficient spaces are independent of T. Their construction is polynomial time by Lemma 2. Interpolation, finding U', solving the affine identity over $\mathbb{F}_q$, intersecting with U', and taking the linear span are finite-field linear algebra of polynomial size. Standard deterministic finite-field algorithms construct the required field representations in time polynomial in q and m. No enumeration of the $q^{O(\ell/\varepsilon^2)}$ vectors in the containing space is needed to output its basis. This completes the proof. □

5 Relation to the cited theorem

For $\ell = 1$ this is the subfield-evaluation Reed–Solomon subcode mechanism of Guruswami and Xing [2]; a recent account appears in [3, Section 5.3]. The list-recovery modification is exactly the imposition of one interpolation equation for every pair (α_i, y) with $y \in T_i$, rather than one per coordinate. Choosing $h = \Theta(\ell/\varepsilon)$ and coefficient-space codimension fraction $\Theta(\varepsilon)$ gives the stated dimension bound. The proof above treats the affine-to-linear conversion and all finite-length exceptions explicitly.

References

[1] V. Guruswami and S. Kopparty. *Explicit subspace designs*. Combinatorica 36 (2016), 161–185. Construction cited as Theorem 7 in the March 2013 preprint. https://www.math.toronto.edu/swastik/subspace-designs.pdf.

[2] V. Guruswami and C. Xing. *Optimal rate list decoding over bounded alphabets using algebraic-geometric codes*. Journal of the ACM 69(2) (2022), Article 10. https://doi.org/10.1145/3506668.

[3] M. Kumar and N. Ron-Zewi. *Advances in List Decoding of Polynomial Codes*. ECCC Report TR26-032 (2026), Section 5.3. https://eccc.weizmann.ac.il/report/2026/032/.